UCL DEPARTMENT OF COMPUTER SCIENCE

COMP0080: Graphical Models

Topic-Grouped Past Examination Questions & Exhaustive Solutions Guide

Categorized from 2021–2022, 2023–2024, and Classic Sample Examination Papers

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Master Course Syllabus & Topic Roadmap

This master revision guide categorizes **all past examination questions** for UCL COMP0080 (Graphical Models) by topic, featuring **each question immediately followed by its exhaustive step-by-step solution**:

Topic Area	Exam Questions & Core Algorithms Covered
Topic 1: Representations	Independence in Joint Tables, Pairwise Independence, Markov Net Separation, State Probabilities, Markov Equivalence, Verma & Pearl Algorithm.
Topic 2: Exact Inference	Factor Graphs, Sum-Product / Belief Propagation, Junction Tree Algorithm (JTA), Moralisation, Triangulation, Running Intersection Property.
Topic 3: Time Series	Hidden Markov Models (HMMs), Forward Filtering (α), Backward Smoothing (β), Likelihood, Prediction, Viterbi Decoding, Exact Sampling.
Topic 4: Learning	Maximum Likelihood Estimation (MLE), Bipartite Markov Nets, Empirical KL Minimization, Randomized Response Surveys, Baum-Welch EM.
Topic 5: Approximate Inference	Matrix KL Divergence, Variational Free Energy ($F(q)$), Mode-Seeking vs Mode-Covering KL, Rejection Sampling, MCMC, Gibbs & MH.
Appendix	Standard Formulas: Chain Rule ($N = 2, 3, \ldots, n$), Bayes' Rule, D-Separation, KL Divergence, Jensen's Inequality, Junction Tree RIP, HMM recursions.

TOPIC 1: Representations & Independence in Graphical Models

Question 1.1: Independence in Joint Probability Tables

[Source: 2021–22 Final Exam, Question 1(a)]
Suppose that X_1 and X_2 are two random variables, where each can take one of three states. The joint distribution $P(X_1 = i, X_2 = j)$ is described by matrix P:

$$P = \begin{pmatrix} 0.06 & 0.37 & 0.12 \\ 0.01 & 0.07 & 0.02 \\ 0.03 & 0.26 & 0.06 \end{pmatrix}$$

1. Are X_1 and X_2 independent? Support your answer. [2 Marks]
2. What is the minimal number of entries in the table that you need to change to make them independent while keeping the table a valid probability distribution (non-negative and summing to 1.0)? State the changed entries and the new matrix. [4 Marks]

Solution 1.1

Core Concept Summary: Random variables X_1 and X_2 are statistically independent iff their joint probability matrix P is a rank-1 outer product of marginal vectors $P = rc^T$ (i.e. $P(i, j) = P(X_1 = i)P(X_2 = j)$ for all i, j), which requires all 2×2 subdeterminants of P to be zero.

1. **Exhaustive Verification of Independence:** Calculate row marginal vector $r = [r_1, r_2, r_3]^T$ by summing across columns, and column marginal vector $c = [c_1, c_2, c_3]^T$ by summing across rows:

$$r_1 = 0.06 + 0.37 + 0.12 = 0.55$$

$$r_2 = 0.01 + 0.07 + 0.02 = 0.10$$

$$r_3 = 0.03 + 0.26 + 0.06 = 0.35$$

$$c_1 = 0.06 + 0.01 + 0.03 = 0.10$$

$$c_2 = 0.37 + 0.07 + 0.26 = 0.70$$

$$c_3 = 0.12 + 0.02 + 0.06 = 0.20$$

Notice total sum $\sum_{i,j} P_{ij} = 0.55 + 0.10 + 0.35 = 1.00$. Now test entry $(1, 1)$:

$$r_1 \times c_1 = 0.55 \times 0.10 = 0.055 \neq 0.060 = P_{11}$$

Since $P_{11} \neq r_1 c_1$, the joint distribution does not factorize. Thus, X_1 and X_2 are **not independent**.

2. **Minimal Entries Modification Proof & Derivation:** An independent probability matrix must be rank 1 ($P = rc^T$). Therefore, every 2×2 submatrix of P must have a determinant equal to 0. Let us evaluate the subdeterminants of matrix P:

$$\det \begin{pmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{pmatrix} = \det \begin{pmatrix} 0.06 & 0.37 \\ 0.01 & 0.07 \end{pmatrix} = (0.06)(0.07) - (0.37)(0.01) = 0.0042 - 0.0037 = 0.0005 \neq 0$$

$$\det \begin{pmatrix} P_{22} & P_{23} \\ P_{32} & P_{33} \end{pmatrix} = \det \begin{pmatrix} 0.07 & 0.02 \\ 0.26 & 0.06 \end{pmatrix} = (0.07)(0.06) - (0.02)(0.26) = 0.0042 - 0.0052 = -0.0010 \neq 0$$

Can we make the matrix rank 1 by changing only **1 single entry**? If we change only entry $P_{22} = x$, setting the first subdeterminant to zero requires $(0.06)(x) - 0.0037 = 0 \implies x = 0.06167$. Setting the second subdeterminant to zero requires $(x)(0.06) - 0.0052 = 0 \implies x = 0.08667$. These requirements conflict, so changing 1 entry is **impossible**.
Therefore, minimal changes required: **2 entries**. Observe column 1 $[0.06, 0.01, 0.03]^T = 0.10 \times [0.6, 0.1, 0.3]^T$ and column 3 $[0.12, 0.02, 0.06]^T = 0.20 \times [0.6, 0.1, 0.3]^T$. Both columns 1 and 3 share the exact same row ratio vector $[0.6, 0.1, 0.3]^T$. Only Column 2 $[0.37, 0.07, 0.26]^T$ violates this ratio. Setting reference $P_{22} = 0.07$ (so $c_2 = 0.70$), we modify Column 2 to match ratio $[0.6, 0.1, 0.3]^T$: - Change $P_{12} = 0.70 \times 0.6 = \mathbf{0.42}$ (was 0.37). - Keep $P_{22} = 0.70 \times 0.1 = \mathbf{0.07}$ (unchanged). - Change $P_{32} = 0.70 \times 0.3 = \mathbf{0.21}$ (was 0.26).

The new matrix is:

$$P_{\text{new}} = \begin{pmatrix} 0.06 & 0.42 & 0.12 \\ 0.01 & 0.07 & 0.02 \\ 0.03 & 0.21 & 0.06 \end{pmatrix} = \begin{pmatrix} 0.60 \\ 0.10 \\ 0.30 \end{pmatrix} \begin{pmatrix} 0.10 & 0.70 & 0.20 \end{pmatrix}$$

All entries are non-negative, sum $\sum_{i,j} P_{\text{new},ij} = 1.00$, and the matrix is exactly rank 1. Minimal number of changed entries is **2** (P_{12} and P_{32}).

Question 1.2: Dependencies and Pairwise Independence (XOR)

[Source: 2021–22 Final Exam, Q1(b-c) & 2023–24 Final Exam, Q1(a)]

1. Suppose $A \not\perp\!\!\!\perp B$ and $B \not\perp\!\!\!\perp C$. Does that necessarily imply $A \not\perp\!\!\!\perp C$? Prove or provide a counterexample. [3 Marks]
2. Give an example of a joint distribution in which $A \perp\!\!\!\perp B$, $B \perp\!\!\!\perp C$, $C \perp\!\!\!\perp A$ (pairwise independence), but $A \not\perp\!\!\!\perp B \mid C$. [3 Marks]

Solution 1.2

Core Concept Summary: Statistical dependence is non-transitive ($A \not\perp\!\!\!\perp B \wedge B \not\perp\!\!\!\perp C \not\Longrightarrow A \not\perp\!\!\!\perp C$). Pairwise independence between all pairs of variables does not guarantee mutual independence or conditional independence given remaining variables.

1. **Dependency Transitivity Proof & Counterexample:** Statistical dependence is **not transitive**. Let A and C be independent fair coin flips: $A \sim \text{Bernoulli}(0.5)$ and $C \sim \text{Bernoulli}(0.5)$ with $A \perp\!\!\!\perp C$. Define variable B as the composite tuple $B = (A, C)$. - Since B deterministically contains A, knowing B gives full information about $A \implies A \not\perp\!\!\!\perp B$. - Since B deterministically contains C, knowing B gives full information about $C \implies B \not\perp\!\!\!\perp C$. - Yet by definition, $A \perp\!\!\!\perp C$. Hence, dependence between A and B, and between B and C, does not imply dependence between A and C.
2. **Pairwise Independence with Conditional Dependence (XOR Gate):** Let $A, B \in \{0, 1\}$ be independent fair binary random variables:

$$P(A=0) = P(A=1) = 0.5, \quad P(B=0) = P(B=1) = 0.5, \quad P(A,B) = P(A)P(B) = 0.25$$

Define $C = A \oplus B$ (XOR sum modulo 2). $C = 1$ if $A \neq B$ and $C = 0$ if $A = B$.
Check pairwise independence for all pairs:

- Pair (A, B): $P(A=1, B=1) = 0.25 = P(A)P(B) \implies A \perp\!\!\!\perp B$.
- Pair (B, C): Note that $C = 1 \iff A \oplus B = 1$. Given $B = 1$, $C = 1 \iff A \oplus 1 = 1 \iff A = 0$. Thus the joint event $\{B = 1, C = 1\}$ is identical to $\{B = 1, A = 0\}$, giving $P(B=1, C=1) = P(B=1, A=0) = 0.25 = P(B)P(C) \implies B \perp\!\!\!\perp C$.
- Pair (A, C): Similarly, given $A = 1$, $C = 1 \iff 1 \oplus B = 1 \iff B = 0$. Thus $P(A=1, C=1) = P(A=1, B=0) = 0.25 = P(A)P(C) \implies A \perp\!\!\!\perp C$.

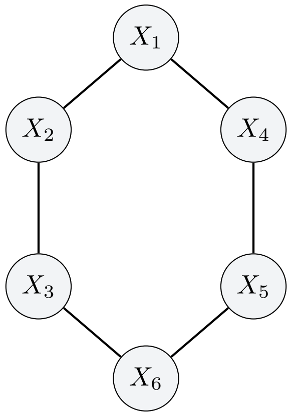
Now evaluate conditional independence $A \perp\!\!\!\perp B \mid C$: Condition on $C = 0$ (which implies $A = B$):

$$P(A=1, B=1 \mid C=0) = \frac{P(A=1, B=1, C=0)}{P(C=0)} = \frac{0.25}{0.50} = 0.50$$

However, $P(A=1 \mid C=0)P(B=1 \mid C=0) = 0.50 \times 0.50 = 0.25$. Since $0.50 \neq 0.25$, conditioning on C makes A and B strongly dependent ($A \not\perp\!\!\!\perp B \mid C$).

Question 1.3: Markov Network Marginal Independence

[Source: 2021–22 Final Exam, Q1(d) & 2023–24 Final Exam, Q1(c)]
Consider the undirected Markov Network on variables $X_1, \ldots, X_6$:



Write down the set of all conditional and marginal independence statements guaranteed to hold for the marginal distribution $P(X_2, X_4, X_6)$. Justify using graph separation. [6 Marks]

Solution 1.3

Core Concept Summary: Marginalizing out variables from an undirected Markov Network connects all neighboring nodes of each eliminated variable, forming a reduced graph where separation determines valid conditional independence statements.

1. **Graph Reduction via Elimination of $\{X_1, X_3, X_5\}$:** To construct the marginal distribution $P(X_2, X_4, X_6)$, we integrate out variables X_1, X_3, X_5:
   - Eliminating X_1 (connected to X_2 and X_4) introduces a direct induced edge (X_2, X_4).
   - Eliminating X_3 (connected to X_2 and X_6) introduces a direct induced edge (X_2, X_6).
   - Eliminating X_5 (connected to X_4 and X_6) introduces a direct induced edge (X_4, X_6).
2. **Marginal Topology & Separation Analysis:** The resulting marginal graph on $\{X_2, X_4, X_6\}$ forms a fully connected 3-clique (K_3) where every node is directly connected to every other node. Because every pair in $\{X_2, X_4, X_6\}$ is adjacent, no node subset can separate any pair of variables. Therefore, the set of guaranteed independence statements for $P(X_2, X_4, X_6)$ is $\emptyset$ **(the empty set)**.

Question 1.4: Exact State Probability in Pairwise Markov Net

[Source: 2023–24 Final Exam, Question 1(b)]
Consider a pairwise Markov Network on 3 binary variables $X_1, X_2, X_3 \in \{0, 1\}$ with joint distribution $P(X) \propto \Phi(X_1, X_2)\Phi(X_2, X_3)\Phi(X_3, X_1)$ where:

$$\Phi(X_i, X_j) = \begin{cases} 2 & \text{if } X_i = X_j \\ 1 & \text{if } X_i \neq X_j \end{cases}$$

Compute the exact probability $P(X_1 = X_2 = X_3)$ of all three variables being in the same state. [5 Marks]

Solution 1.4

Core Concept Summary: In a Markov network, the joint probability of state X is $P(X) = \frac{1}{Z}\prod_c \Phi_c(X_c)$ where partition function $Z = \sum_{X'}\prod_c \Phi_c(X')$ normalizes unnormalized potential weights across all state configurations.

1. **Exhaustive Table of State Potential Weights:** The state space contains $2^3 = 8$ configurations. Unnormalized weight $W(X) = \Phi(X_1, X_2)\Phi(X_2, X_3)\Phi(X_3, X_1)$:
2. **Partition Function & Exact Probability Computation:** - Weight sum of equal states: $W_{\text{equal}} = 8 + 8 = 16$. - Weight sum of unequal states: $W_{\text{unequal}} = 6 \times 2 = 12$. - Partition Function: $Z = W_{\text{equal}} + W_{\text{unequal}} = 16 + 12 = 28$.
   Exact probability of all three variables being equal:

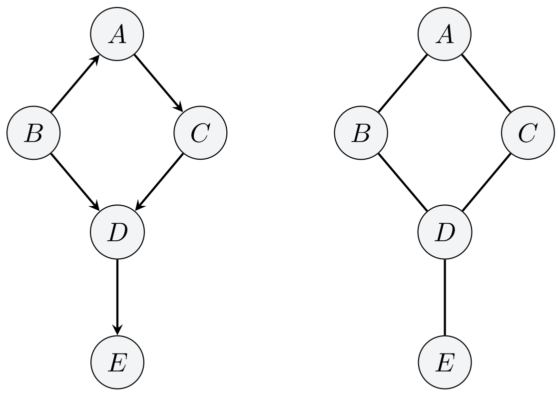
$$P(X_1 = X_2 = X_3) = \frac{W_{\text{equal}}}{Z} = \frac{16}{28} = \mathbf{\frac{4}{7}} \approx 0.5714$$

State (X_1, X_2, X_3)	$\Phi(X_1, X_2)$	$\Phi(X_2, X_3)$	$\Phi(X_3, X_1)$	Unnormalized Weight $W(X)$	Classification
$(0, 0, 0)$	2	2	2	$2 \times 2 \times 2 = 8$	Equal States
$(1, 1, 1)$	2	2	2	$2 \times 2 \times 2 = 8$	Equal States
$(0, 0, 1)$	2	1	1	$2 \times 1 \times 1 = 2$	Unequal States
$(0, 1, 0)$	1	1	2	$1 \times 1 \times 2 = 2$	Unequal States
$(1, 0, 0)$	1	2	1	$1 \times 2 \times 1 = 2$	Unequal States
$(0, 1, 1)$	1	2	1	$1 \times 2 \times 1 = 2$	Unequal States
$(1, 0, 1)$	1	1	2	$1 \times 1 \times 2 = 2$	Unequal States
$(1, 1, 0)$	2	1	1	$2 \times 1 \times 1 = 2$	Unequal States

Question 1.5: Markov Equivalence & Graph Comparisons

[Source: 2021–22 Final Exam, Q1(e) & 2023–24 Final Exam, Q1(d)]

1. Compare Model 1 (DAG) and Model 2 (Markov Network):



Model 1 (DAG) **Model 2 (Markov Network)**

Do they encode the same set of independence assumptions? List all differences. [6 Marks]

2. Compare G1 ($A \to C \leftarrow B$), G2 ($A \leftarrow C \to B$), G3 ($A \to C \to B$), G4 ($A \leftarrow C \leftarrow B$). Which are Markov equivalent? [6 Marks]

Solution 1.5

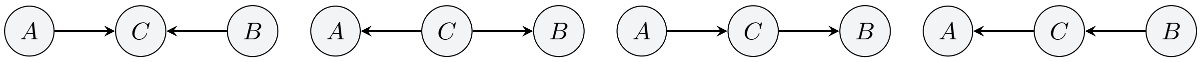
Core Concept Summary: Two graphs are Markov Equivalent if and only if they encode the exact same set of conditional independence statements. For DAGs, this requires having identical undirected skeletons and identical immoralities (v-structures $X \to Z \leftarrow Y$ with no edge between X and Y).

1. **Model 1 (DAG) vs Model 2 (Markov Net) Independence Differences:** Both models share the exact same undirected skeleton: $\{(A, B), (A, C), (B, D), (C, D), (D, E)\}$. However, Model 1 contains an immorality at D ($B \to D \leftarrow C$ with no edge between B and C).
   Conditional independence comparison between Model 1 and Model 2:
   - **Statement** $B \perp\!\!\!\perp C \mid A$: - **Model 1 (DAG)**: Path $B \to D \leftarrow C$ is blocked at collider D (since D is not conditioned on). Path $B \leftarrow A \to C$ is blocked by conditioning on A. Thus $B \perp\!\!\!\perp C \mid A$ is **TRUE**. - **Model 2 (Markov Net)**: Conditioning on A does not block path $B - D - C$. Graph separation fails. Thus $B \perp\!\!\!\perp C \mid A$ is **FALSE**.
   - **Statement** $B \perp\!\!\!\perp C \mid \{A, D\}$: - **Model 1 (DAG)**: There are two paths between B and C: 1. Path $B \leftarrow A \to C$: Blocked because non-collider A is conditioned on ($A \in \{A, D\}$). 2. Path $B \to D \leftarrow C$: Here D is a **collider** ($B \to D \leftarrow C$). Under D-separation rules, conditioning on a collider D ($D \in \{A, D\}$) **opens (unblocks)** the path. Information flows between B and C through D ("explaining away"). Because Path 2 is open, graph d-separation fails. Thus $B \perp\!\!\!\perp C \mid \{A, D\}$ is **FALSE**. - **Model 2 (Markov Net)**: In undirected graphs, conditional independence is standard node separation (no colliders exist). Observing node set $\{A, D\}$ blocks both undirected Path 1 ($B - A - C$) and undirected Path 2 ($B - D - C$). Every path between B and C is blocked. Thus $B \perp\!\!\!\perp C \mid \{A, D\}$ is **TRUE**.

Since their conditional independence sets differ, Model 1 and Model 2 are **not Markov Equivalent**.

2. **3-Node DAG Markov Equivalence Evaluation:** According to the **Verma & Pearl Theorem**, two DAGs are Markov equivalent if and only if:
   (i) They share the exact same undirected skeleton.
   (ii) They contain the exact same set of immoralities (v-structures $X \to Z \leftarrow Y$ where X and Y are not adjacent).

   Visual Topology Comparison of G_1, G_2, G_3, G_4:

   G_1**: Immorality** G_2**: Fork** G_3**: Chain** G_4**: Reverse Chain**

   Exhaustive Graph-by-Graph Evaluation:

   - **Skeleton Verification**: Stripping arrowheads from G_1, G_2, G_3, G_4 yields the exact same undirected skeleton $A - C - B$ with edge set $\{(A, C), (C, B)\}$ and no edge (A, B).
   - **Immorality / V-Structure Check**:
     – G_1 ($A \to C \leftarrow B$): Node C is a collider with non-adjacent parents A and B. Contains immorality $\{A \to C \leftarrow B\}$. - Unconditioned: Path $A \to C \leftarrow B$ is blocked at collider $C \implies A \perp\!\!\!\perp B$. - Conditioned on C: Observing collider C opens path $\implies A \not\perp\!\!\!\perp B \mid C$.
     – G_2 ($A \leftarrow C \to B$): Node C is a fork (common parent). No immorality ($\emptyset$). - Unconditioned: Path open $\implies A \not\perp\!\!\!\perp B$. - Conditioned on C: Observing C blocks path $\implies A \perp\!\!\!\perp B \mid C$.
     – G_3 ($A \to C \to B$): Causal chain. No immorality ($\emptyset$). - Unconditioned: Path open $\implies A \not\perp\!\!\!\perp B$. - Conditioned on C: Observing C blocks path $\implies A \perp\!\!\!\perp B \mid C$.
     – G_4 ($A \leftarrow C \leftarrow B$): Reverse causal chain. No immorality ($\emptyset$). - Unconditioned: Path open $\implies A \not\perp\!\!\!\perp B$. - Conditioned on C: Observing C blocks path $\implies A \perp\!\!\!\perp B \mid C$.

   Markov Equivalence Class Partition: - **Equivalence Class 1** $\{G_2, G_3, G_4\}$: Graphs G_2, G_3, G_4 share identical skeleton $A - C - B$ and identical empty immorality set ($\emptyset$). They encode the exact same conditional independence statement ($A \perp\!\!\!\perp B \mid C$). By Bayes' rule, their joint factorizations are algebraically equivalent:

   $$p(a, b, c) = p(c)p(a \mid c)p(b \mid c) = p(a)p(c \mid a)p(b \mid c) = p(b)p(c \mid b)p(a \mid c)$$

   Therefore, G_2, G_3**, and** G_4 **are Markov Equivalent**. - **Equivalence Class 2** $\{G_1\}$: Graph G_1 contains a unique immorality at C, encoding marginal independence $A \perp\!\!\!\perp B$ with factorization $p(a, b, c) = p(a)p(b)p(c \mid a, b)$. It forms its own distinct equivalence class.

Question 1.6: Fundamental Definitions & Universal Factorizations

[Source: Classic Sample Exam, Question 1(a-b)]

1. Explain what dependency between discrete random variables X_i and X_j means mathematically. [2 Marks]
2. Define a Belief Network. Why must its graphical structure be a Directed Acyclic Graph (DAG)? [2 Marks]
3. Prove why any joint distribution $p(x_1, \ldots, x_N)$ can be written as a Belief Network. [2 Marks]
4. Explain why the lack of an edge in a DAG implies an independence statement, whereas the presence of an edge does not guarantee dependence. [3 Marks]

Solution 1.6

Core Concept Summary: Belief Networks parameterize joint distributions via recursive conditional probability tables (CPTs) over DAG topologies. DAG acyclicity ensures valid probabilistic conditioning without circular dependence.

1. **Mathematical Dependency Definition:** Variables X_i and X_j are dependent if their joint distribution cannot be factorized into the product of their marginals:

$$p(x_i, x_j) \neq p(x_i)p(x_j) \iff p(x_i \mid x_j) \neq p(x_i)$$

2. **Belief Network Definition & DAG Acyclicity Requirement:** A Belief Network (Bayesian Network) is a directed graphical model that factorizes a joint distribution as:

$$p(x_1, \ldots, x_N) = \prod_{i=1}^{N} p(x_i \mid \mathrm{pa}(x_i))$$

The graph must be a DAG to guarantee that the factorized product represents a valid, normalized probability distribution. Directed cycles (e.g. $A \to B \to A$) create circular conditional definitions $p(a \mid b)p(b \mid a)$, which generally do not integrate or sum to 1.

3. **Universal Factorization Proof via Chain Rule:** By the chain rule of probability, any joint distribution over N variables can be expanded without loss of generality as:

$$p(x_1, x_2, \ldots, x_N) = p(x_1)p(x_2 \mid x_1)p(x_3 \mid x_1, x_2) \ldots p(x_N \mid x_1, \ldots, x_{N-1})$$

This expansion directly matches a fully connected DAG where node x_i has parents $\mathrm{pa}(x_i) = \{x_1, \ldots, x_{i-1}\}$. Hence, any joint distribution can be represented as a Belief Network.

4. **Lack of Edge vs Presence of Edge: - Lack of Edge**: Omitting an edge $X_i \to X_j$ asserts a structural independence constraint ($X_j \perp\!\!\!\perp X_i \mid \mathrm{pa}(X_j) \setminus \{X_i\}$) that must hold for *all* parameter choices of CPTs. - **Presence of Edge**: An edge $X_i \to X_j$ allows conditional dependence, but does not guarantee it. For specific numerical choices of CPT parameters (e.g. uniform CPT $p(x_j \mid x_i) = \text{const}$), X_j may remain independent of X_i.

Question 1.7: DAG to Undirected Conversion Counterexample

[Source: Classic Sample Exam, Question 1(b.iii)]
Let DAG A have adjacency matrix A_{ij}. Form undirected adjacency matrix $B_{ij} = A_{ij} \vee A_{ji}$. Can every Belief Network with DAG A be represented by a Markov Network with adjacency B? Explain or give counterexample. [3 Marks]

Solution 1.7

Core Concept Summary: Converting a DAG to a Markov Network by simply stripping arrowheads (without moralizing unmarried parents) fails for collider structures $A \to B \leftarrow C$, because the collider induces marginal independence ($A \perp\!\!\!\perp C$) which is violated in the un-moralized undirected graph.

No. Consider the collider DAG $A \to B \leftarrow C$. - In the DAG, A and C are marginally independent ($A \perp\!\!\!\perp C$). - Stripping arrowheads yields undirected skeleton B with edges (A, B) and (B, C), but no edge (A, C). - In undirected graph B, path $A - B - C$ connects A and C, implying $A \not\!\perp\!\!\!\perp C$. Thus undirected graph B cannot represent the DAG without adding moral edge (A, C).

Question 1.8: Verma & Pearl Markov Equivalence Algorithm

[Source: Classic Sample Exam, Question 1(c)]

1. Define Markov Equivalence between two graphs. [2 Marks]
2. State an algorithm to determine if two DAGs are Markov equivalent. [2 Marks]
3. Check equivalence for Pair 1: $(a \to b \to d, a \to c \to d)$ vs $(a \leftarrow b \to d, a \to c \to d)$. [3 Marks]
4. Check equivalence for Pair 2: $(a \to b \to d, a \to c \to d)$ vs $(a \to b \to d, a \to c \leftarrow d)$. [3 Marks]

Solution 1.8

Core Concept Summary: Verma & Pearl Theorem: Two DAGs are Markov Equivalent if and only if they have the exact same undirected skeleton and the exact same set of immoralities.

1. **Markov Equivalence Definition:** Two graphs G_1 and G_2 are Markov Equivalent if they encode identical sets of conditional independence statements.
2. **Verma & Pearl Algorithm:**
   (i) Construct the undirected skeleton for both DAGs (strip arrowheads). If skeletons differ, return **Not Equivalent**.
   (ii) Identify all immoralities (unmarried colliders $X \to Z \leftarrow Y$ with no edge between X and Y). If the sets of immoralities differ, return **Not Equivalent**. Otherwise return **Markov Equivalent**.
3. **Pair 1 Evaluation:** Both DAGs have skeleton $\{(a,b),(a,c),(b,d),(c,d)\}$ and 0 immoralities. **Markov Equivalent**.
4. **Pair 2 Evaluation:** DAG 1 has no immoralities. DAG 2 has immorality at d ($c \to d \leftarrow b$). **Not Markov Equivalent**.

TOPIC 2: Exact Inference & Factor Graphs

Question 2.1: The Junction Tree Algorithm

[Source: 2021–22 Final Exam, Q2.2 & 2023–24 Final Exam, Q2(b)]
Consider the joint distribution that factorizes as:

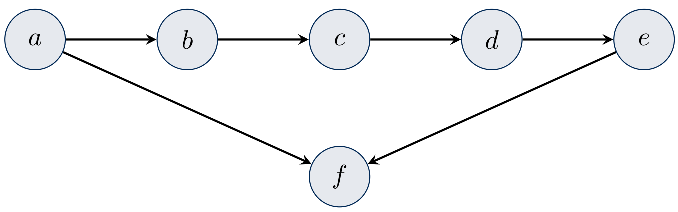
$$p(a, b, c, d, e, f) = p(a)p(b \mid a)p(c \mid b) \\ \times p(d \mid c)p(e \mid d)p(f \mid a, e)$$

1. Draw the DAG representing this distribution. [3 Marks]
2. Draw the moralised version of this graph. Explain parent marrying. [3 Marks]
3. Draw the triangulated version of the graph with the smallest possible maximum clique size. [3 Marks]
4. Construct a Junction Tree and verify the Running Intersection Property (RIP). [3 Marks]
5. Describe potential initialization, absorption, and message schedule. [3 Marks]

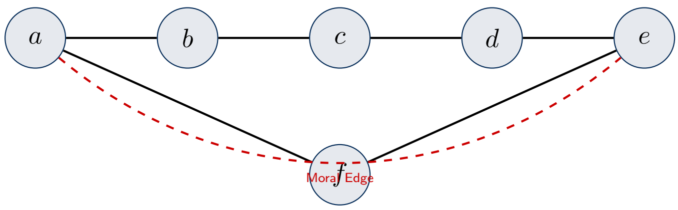
Solution 2.1

Core Concept Summary: The Junction Tree Algorithm converts arbitrary DAGs/Markov networks into a tree of maximal cliques. Initial clique potentials $\psi(C_i)$ are formed by assigning each DAG factor to exactly one clique, ensuring $\prod_i \psi(C_i) \equiv p(x)$. Message absorption preserves joint factorization invariant $p(x) = \frac{\prod \psi(C_i)^*}{\prod \phi(S_{ij})^*}$ while computing exact marginals in $O(N)$ time.

1. **DAG Construction (Directed Graph):**

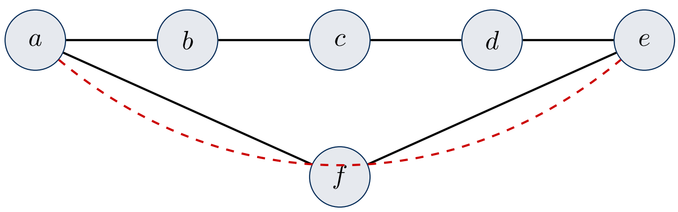


2. **Moralisation & Parent Marrying:** Marry unmarried parents of f (a and e) with a moral edge (a, e), and strip all arrowheads:

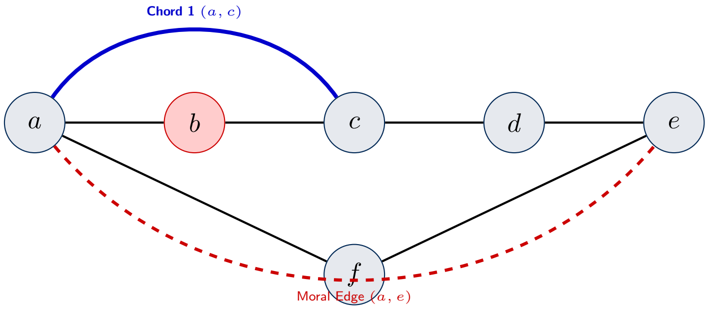


3. **Triangulation Transformations (Step-by-Step Node & Edge Graphs):** Triangulation eliminates cycles of length ≥ 4 without chords by adding edges between non-adjacent neighbors of eliminated nodes during Variable Elimination:

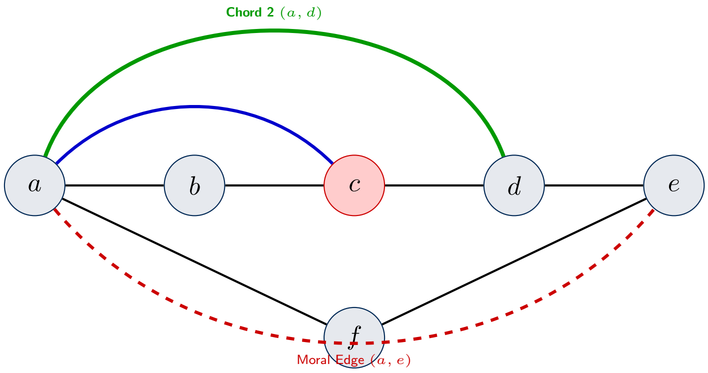
Stage 1: Moral Graph with Chordless 5-Cycle The moralized graph contains an un-triangulated 5-cycle $a - b - c - d - e - a$:



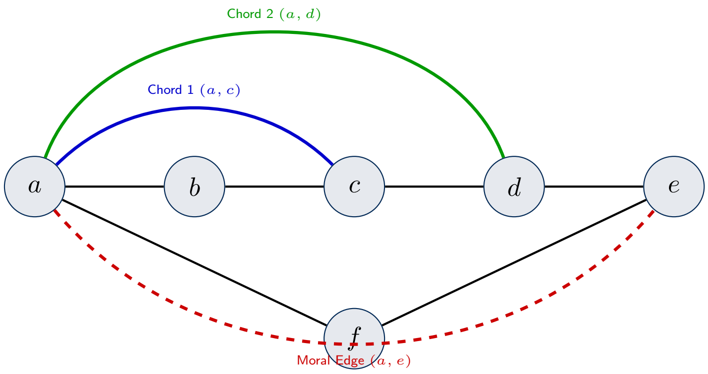
Stage 2: Eliminate Node $b \implies$ Add Chord 1 (a, c) (Curved Arc ABOVE) Eliminating b (whose neighbors are a and c) connects a and c with **Chord 1** (a, c), drawn as a curved arc **ABOVE** node b, forming 3-clique $C_1 = \{a, b, c\}$ and leaving a 4-cycle $a - c - d - e - a$:



Stage 3: Eliminate Node $c \implies$ Add Chord 2 (a, d) (Higher Curved Arc ABOVE) Eliminating c (whose neighbors are a and d) connects a and d with **Chord 2** (a, d), drawn as a higher curved arc **ABOVE** nodes b and c, forming 3-clique $C_2 = \{a, c, d\}$:



Stage 4: Final Triangulated Chordal Graph Showing all curved top chords and bottom moral edge with clear joints:



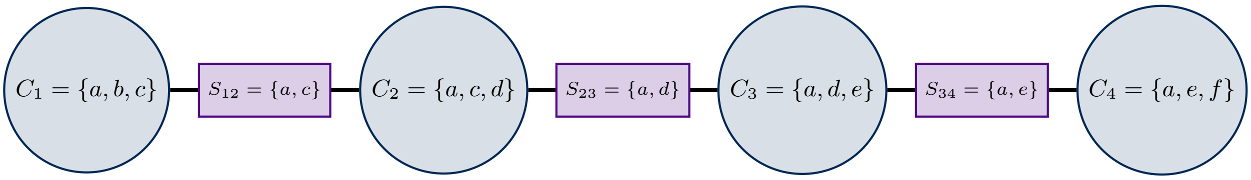
The maximal cliques are: $C_1 = \{a, b, c\}$, $C_2 = \{a, c, d\}$, $C_3 = \{a, d, e\}$, $C_4 = \{a, e, f\}$ (maximum clique size = 3).

4. **Junction Tree Diagram & Running Intersection Property (RIP) Verification:** Using the maximal cliques $\{C_1, C_2, C_3, C_4\}$ derived in 2.1.3, we construct the Junction Tree graph.

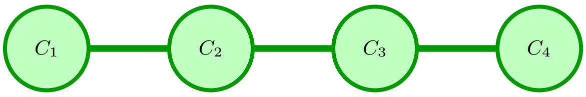
   What are S_{ij} (Separator Sets)? In a Junction Tree, the edges between adjacent clique nodes C_i and C_j contain **Separator Sets** S_{ij}:

- **Mathematical Definition**: S_{ij} is the set intersection of variables shared by clique C_i and clique C_j:
$$S_{ij} = C_i \cap C_j$$
- **Explicit Calculations**:
  - $S_{12} = C_1 \cap C_2 = \{a, b, c\} \cap \{a, c, d\} = \{\mathbf{a}, \mathbf{c}\}$
  - $S_{23} = C_2 \cap C_3 = \{a, c, d\} \cap \{a, d, e\} = \{\mathbf{a}, \mathbf{d}\}$
  - $S_{34} = C_3 \cap C_4 = \{a, d, e\} \cap \{a, e, f\} = \{\mathbf{a}, \mathbf{e}\}$
- **Purpose in Algorithm**: S_{ij} defines the exact variable interface over which messages $\mu_{i\to j}(S_{ij})$ pass between cliques. Dividing by separator potentials $\phi(S_{ij})$ in $p(x) = \frac{\prod_i \psi(C_i)}{\prod_{ij} \phi(S_{ij})}$ prevents double-counting shared variables!

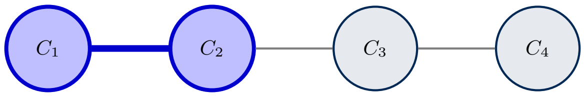
Visual Junction Tree Architecture with Separators:



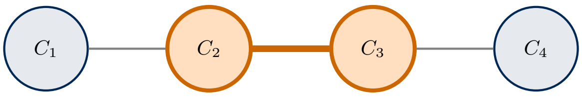
Diagrammatic Verification of Running Intersection Property (RIP): RIP mandates that for every variable x, the set of cliques containing x forms a **single, contiguous connected sub-tree** of the Junction Tree:



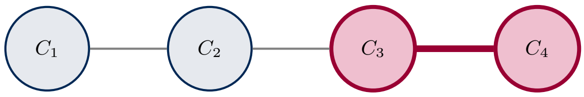
Variable a Sub-tree (Contiguous $C_1 - C_2 - C_3 - C_4$)



Variable c Sub-tree (Contiguous $C_1 - C_2$)



Variable d Sub-tree (Contiguous $C_2 - C_3$)



Variable e Sub-tree (Contiguous $C_3 - C_4$)

Since every shared variable forms a fully connected, contiguous sub-graph on the Junction Tree, the **Running Intersection Property holds strictly**.

5. **Potential Initialization & Message Passing Schedule:**

Derivation of Initial Clique Potential Assignments: The original DAG factorization given in the question is:

$$p(a, b, c, d, e, f) = p(a) \cdot p(b \mid a) \cdot p(c \mid b) \cdot p(d \mid c) \cdot p(e \mid d) \cdot p(f \mid a, e)$$

To initialize a Junction Tree, every CPT factor $p(x_i \mid \mathrm{pa}(x_i))$ from the original DAG must be assigned to **EXACTLY ONE** clique C_k that contains all of its variables $\{x_i\} \cup \mathrm{pa}(x_i)$:

(i) **Initialization of $\psi(C_1)$ for Clique $C_1 = \{a, b, c\}$**: - $p(a)$ has variables $\{a\} \subseteq C_1$. Assign to C_1. - $p(b \mid a)$ has variables $\{a, b\} \subseteq C_1$. Assign to C_1. - $p(c \mid b)$ has variables $\{b, c\} \subseteq C_1$. Assign to C_1.
$$\implies \psi(\mathbf{C_1}) = \mathbf{p(a)\, p(b \mid a)\, p(c \mid b)}$$

(ii) **Initialization of $\psi(C_2)$ for Clique $C_2 = \{a, c, d\}$**: - $p(d \mid c)$ has variables $\{c, d\} \subseteq C_2$. Assign to C_2. - **Why does $\psi(C_2)$ ONLY contain $p(d \mid c)$ and not a?** Because factors $p(a), p(b|a), p(c|b)$ were already assigned to C_1, and no other unassigned factor contains $\{c, d\}$. Although C_2 contains variables $\{a, c, d\}$, $p(d \mid c)$ is uniform/constant with respect to a.

$$\implies \psi(\mathbf{C_2}) = \mathbf{p(d \mid c)}$$

(iii) **Initialization of $\psi(C_3)$ for Clique $C_3 = \{a, d, e\}$:** - $p(e \mid d)$ has variables $\{d, e\} \subseteq C_3$. Assign to C_3.

$$\implies \psi(\mathbf{C_3}) = \mathbf{p(e \mid d)}$$

(iv) **Initialization of $\psi(C_4)$ for Clique $C_4 = \{a, e, f\}$:** - $p(f \mid a, e)$ has variables $\{a, e, f\} \subseteq C_4$. Assign to C_4.

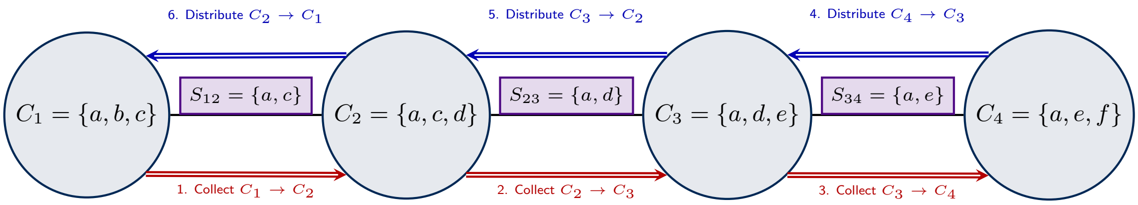
$$\implies \psi(\mathbf{C_4}) = \mathbf{p(f \mid a, e)}$$

Verification of Joint Product Equivalence: All separators are initialized to unity ($\phi(S_{12}) = \phi(S_{23}) = \phi(S_{34}) = 1$). Multiplying all initial clique potentials together:

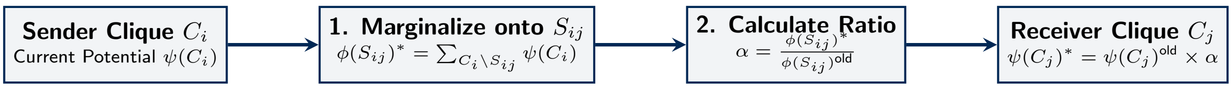
$$\prod_{i=1}^{4} \psi(C_i) = [p(a)p(b|a)p(c|b)] \cdot [p(d|c)] \cdot [p(e|d)] \cdot [p(f|a, e)] \equiv p(a, b, c, d, e, f)$$

This matches the original joint probability distribution exactly without any missing or duplicated factors!

Diagrammatic Junction Tree Message Absorption Schedule Message passing runs in two sequential phases: **Collect Pass** (inward from leaves to root) followed by **Distribute Pass** (outward from root to leaves).



The Message Absorption Update Mechanism (Sender C_i → Receiver C_j): Whenever clique C_i passes a message to neighboring clique C_j across separator S_{ij}, two exact operations occur:



Complete Step-by-Step Algebraic Trace of Absorption Equations:

(i) **Pass 1 (Collect Phase towards Root C_4):**

- **Step 1 ($C_1 \to C_2$ across $S_{12} = \{a, c\}$):**

$$\phi(S_{12})^* = \sum_b \psi(C_1) = \sum_b p(a)p(b|a)p(c|b) = p(a, c)$$

Update C_2: $\psi(C_2)^* = \psi(C_2) \cdot \frac{\phi(S_{12})^*}{1} = p(d|c) \cdot p(a, c) = \mathbf{p(a, c, d)}$.

- **Step 2 ($C_2 \to C_3$ across $S_{23} = \{a, d\}$):**

$$\phi(S_{23})^* = \sum_c \psi(C_2)^* = \sum_c p(a, c, d) = p(a, d)$$

Update C_3: $\psi(C_3)^* = \psi(C_3) \cdot \frac{\phi(S_{23})^*}{1} = p(e|d) \cdot p(a, d) = \mathbf{p(a, d, e)}$.

- **Step 3 ($C_3 \to C_4$ across $S_{34} = \{a, e\}$):**

$$\phi(S_{34})^* = \sum_d \psi(C_3)^* = \sum_d p(a, d, e) = p(a, e)$$

Update C_4: $\psi(C_4)^* = \psi(C_4) \cdot \frac{\phi(S_{34})^*}{1} = p(f|a, e) \cdot p(a, e) = \mathbf{p(a, e, f)}$.

At the end of the Collect Phase, root clique C_4 holds the true exact marginal $p(a, e, f)$!

(ii) **Pass 2 (Distribute Phase from Root C_4 back to C_1):**
- **Step 4 ($C_4 \to C_3$ across $S_{34} = \{a, e\}$):**
$$\phi(S_{34})^{**} = \sum_f \psi(C_4)^* = \sum_f p(a, e, f) = p(a, e)$$
Update C_3: $\psi(C_3)^{**} = \psi(C_3)^* \cdot \frac{\phi(S_{34})^{**}}{\phi(S_{34})^*} = p(a, d, e) \cdot \frac{p(a,e)}{p(a,e)} = \mathbf{p(a, d, e)}$.
- **Step 5 ($C_3 \to C_2$ across $S_{23} = \{a, d\}$):**
$$\phi(S_{23})^{**} = \sum_e \psi(C_3)^{**} = \sum_e p(a, d, e) = p(a, d)$$
Update C_2: $\psi(C_2)^{**} = \psi(C_2)^* \cdot \frac{\phi(S_{23})^{**}}{\phi(S_{23})^*} = p(a, c, d) \cdot \frac{p(a,d)}{p(a,d)} = \mathbf{p(a, c, d)}$.
- **Step 6 ($C_2 \to C_1$ across $S_{12} = \{a, c\}$):**
$$\phi(S_{12})^{**} = \sum_d \psi(C_2)^{**} = \sum_d p(a, c, d) = p(a, c)$$
Update C_1: $\psi(C_1)^* = \psi(C_1) \cdot \frac{\phi(S_{12})^{**}}{\phi(S_{12})^*} = [p(a)p(b|a)p(c|b)] \cdot \frac{p(a,c)}{p(a,c)} = \mathbf{p(a, b, c)}$.

Upon convergence, every clique potential $\psi(C_i)^*$ equals the true marginal $p(C_i)$ of its variable set!

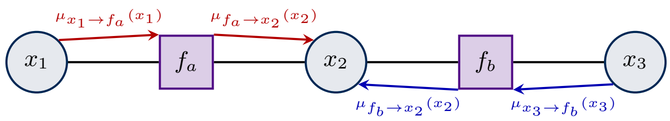
Question 2.2: Sum-Product Algorithm on Factor Graphs

[Source: Classic Sample Exam, Question 2(a)]
Describe the sum-product algorithm for computing marginals in a factor graph. Explain which factor graph topologies the algorithm is applicable to. [4 Marks]

Solution 2.2

Core Concept Summary: The Sum-Product algorithm computes exact marginals on tree-structured factor graphs via two-way message passing between variable nodes x_i and factor nodes f_a.

Diagrammatic Factor Graph Message Passing



Sum-Product message update equations:
$$\mu_{f \to i}(x_i) = \sum_{\chi_f \setminus \{i\}} f(x_i, \chi_f) \prod_{j \in \text{ne}(f) \setminus \{i\}} \mu_{j \to f}(x_j)$$
$$\mu_{i \to f}(x_i) = \prod_{g \in \text{ne}(i) \setminus \{f\}} \mu_{g \to i}(x_i)$$

Applicable for exact marginals on single-tree factor graphs (no loops).

TOPIC 3: Time Series & Hidden Markov Models (HMMs)

Question 3.1: HMM Posterior Graph & Exact Sampling

[Source: 2021–22 Final Exam, Q2(b.i-ii) & Classic Sample Exam, Q2(b.vii)]

1. When sequence $v_{1:T}$ is observed, it induces posterior distribution $p_v(h_{1:T}) = p(h_{1:T} \mid v_{1:T})$. What is the graphical model of this posterior? Is $h_1 \perp\!\!\!\perp h_T$? [3 Marks]
2. Can we efficiently sample from $p_v(h_{1:T})$ exactly? Describe how. [3 Marks]

Solution 3.1

Core Concept Summary: Conditioning an HMM on observations $v_{1:T}$ induces an undirected linear chain posterior over hidden states $h_{1:T}$ where h_1 and h_T remain dependent.

1. **Posterior Topology & Dependency:** Linear chain graph $h_1 - h_2 - \cdots - h_T$. h_1 and h_T are **dependent**.
2. **Forward-Filtering Backward-Sampling Algorithm:**
   (i) Compute forward messages $\alpha_t(h_t) = p(h_t, v_{1:t})$ for $t = 1 \dots T$.
   (ii) Sample $h_T^* \sim p(h_T \mid v_{1:T}) \propto \alpha_T(h_T)$.
   (iii) Step backward for $t = T-1 \dots 1$: sample $h_t^* \sim p(h_t \mid h_{t+1}^*, v_{1:t}) \propto p(h_{t+1}^* \mid h_t)\alpha_t(h_t)$.

Question 3.2: Deterministic State Decoupling & Step-by-Step Filtering

[Source: 2021–22 Final Exam, Question 2.3]

Given $v_t \in \{0,1,2\}$, $h_t \in \{0,1\}$, $p(h_1 = 1) = 0.5$, transition matrix $T = \begin{pmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{pmatrix}$, emission matrix $B = \begin{pmatrix} 0.3 & 0.5 \\ 0.6 & 0 \\ 0.1 & 0.5 \end{pmatrix}$, and sequence $v_{1:10} = [0,1,0,2,0,2,1,0,2,0]$:

1. Is $h_1 \perp\!\!\!\perp h_{10} \mid v_{1:10}$ for this specific sequence? Why? [2 Marks]
2. Compute exact filtering distributions $p(h_8 \mid v_{1:8})$ and $p(h_9 \mid v_{1:9})$. [3 Marks]

Solution 3.2

Core Concept Summary: Zero-emission probabilities ($B_{v,h} = 0$) pin hidden states deterministically, acting as hard observed split nodes that d-separate past and future hidden states in the chain.

1. **Deterministic State Decoupling Analysis:** Emission row 1 for state $h = 1$ is $B_{1,1} = p(v = 1 \mid h = 1) = 0$. Since $v_2 = 1$ and $v_7 = 1$, states are pinned deterministically: $h_2 = 0$ and $h_7 = 0$ with probability 1. Conditioning on $h_7 = 0$ d-separates h_1 from h_{10}. They are **independent**.
2. **Filtering Calculations for $t = 8, 9$:**
   (a) Base at $t = 7$: $p(h_7 \mid v_{1:7}) = \begin{pmatrix} 1.0 \\ 0.0 \end{pmatrix}$.
   (b) Predict $t = 8$: $p(h_8 \mid v_{1:7}) = Tp(h_7 \mid v_{1:7}) = \begin{pmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{pmatrix} \begin{pmatrix} 1.0 \\ 0.0 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$.
   (c) Observe $v_8 = 0$: Unnormalized $\begin{pmatrix} 0.3 \times 0.5 \\ 0.5 \times 0.5 \end{pmatrix} = \begin{pmatrix} 0.15 \\ 0.25 \end{pmatrix} \implies p(h_8 \mid v_{1:8}) = \begin{pmatrix} 0.375 \\ 0.625 \end{pmatrix} = \begin{pmatrix} 3/8 \\ 5/8 \end{pmatrix}$.
   (d) Predict $t = 9$: $p(h_9 \mid v_{1:8}) = Tp(h_8 \mid v_{1:8}) = \begin{pmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{pmatrix} \begin{pmatrix} 0.375 \\ 0.625 \end{pmatrix} = \begin{pmatrix} 0.6875 \\ 0.3125 \end{pmatrix}$.
   (e) Observe $v_9 = 2$: Unnormalized $\begin{pmatrix} 0.1 \times 0.6875 \\ 0.5 \times 0.3125 \end{pmatrix} = \begin{pmatrix} 0.06875 \\ 0.15625 \end{pmatrix} \implies p(h_9 \mid v_{1:9}) = \begin{pmatrix} 11/36 \\ 25/36 \end{pmatrix} \approx \begin{pmatrix} \mathbf{0.3056} \\ \mathbf{0.6944} \end{pmatrix}$.

Question 3.3: HMM Filtering, Smoothing & Viterbi Recursions

[Source: Classic Sample Exam, Question 2(b.i-vi)]

1. Derive forward filtering $\alpha_t(h_t) = p(h_t, v_{1:t})$. [3 Marks]
2. Derive backward smoothing $\beta_t(h_t) = p(v_{t+1:T} \mid h_t)$. [3 Marks]
3. Compute sequence likelihood $p(v_{1:T})$. [5 Marks]
4. Derive prediction $p(v_{T+1} \mid v_{1:T})$. [5 Marks]
5. Derive Viterbi algorithm for $\text{argmax}_{h_{1:T}} p(h_{1:T} \mid v_{1:T})$. [3 Marks]

Solution 3.3

Core Concept Summary: HMM dynamic programming passes forward messages α_t and backward messages β_t to compute marginal posteriors, likelihoods, and Viterbi MAP state paths.

1. **Forward Filtering $\alpha_t(h_t)$:** $\alpha_t(h_t) = p(v_t \mid h_t) \sum_{h_{t-1}} p(h_t \mid h_{t-1}) \alpha_{t-1}(h_{t-1})$.
2. **Backward Smoothing $\beta_t(h_t)$:** $\beta_t(h_t) = \sum_{h_{t+1}} p(v_{t+1} \mid h_{t+1}) p(h_{t+1} \mid h_t) \beta_{t+1}(h_{t+1})$.
3. **Sequence Likelihood:** $p(v_{1:T}) = \sum_{h_T} \alpha_T(h_T)$.
4. **One-step Ahead Prediction:** $p(v_{T+1} \mid v_{1:T}) = \sum_{h_{T+1}} p(v_{T+1} \mid h_{T+1}) \sum_{h_T} p(h_{T+1} \mid h_T) p(h_T \mid v_{1:T})$.
5. **Viterbi Max-Product Recursion:** $V_t(h_t) = p(v_t \mid h_t) \max_{h_{t-1}} [p(h_t \mid h_{t-1}) V_{t-1}(h_{t-1})]$.

TOPIC 4: Parameter Learning & EM Algorithm

Question 4.1: Bipartite Disease-Symptom Markov Network

[Source: 2021–22 Final Exam, Question 2.1]
Given $p(s,d) = \frac{1}{Z}\exp(s^T W d + a^T s + b^T d)$:

1. Draw the Markov Network. Describe structure. [3 Marks]
2. Derive $p(s \mid d)$. Is it factorized? [4 Marks]
3. Derive log-likelihood L for N records (s^n, d^n). [3 Marks]
4. Derive gradients $\frac{\partial L}{\partial W_{ij}}$, $\frac{\partial L}{\partial a_i}$, $\frac{\partial L}{\partial b_j}$. [5 Marks]
5. Efficiency in general case? [3 Marks]

Solution 4.1

Core Concept Summary: Conditioning a bipartite Markov network on one layer (d) breaks all horizontal ties, causing the conditional distribution over symptoms $p(s \mid d)$ to factorize completely into independent Bernoulli distributions.

1. **Graph Structure:** Bipartite graph between s_i and d_j.
2. **Conditional Distribution $p(s \mid d)$:** $p(s \mid d) = \prod_{i=1}^{S} \sigma(W_i.d + a_i)^{s_i}(1 - \sigma(W_i.d + a_i))^{1-s_i}$. Fully factorized.
3. **Log-Likelihood Formulation:** $L = \sum_{n=1}^{N}\left[(s^n)^T W d^n + a^T s^n + b^T d^n\right] - N \log Z(W, a, b)$.
4. **Gradient Derivations:** $\frac{\partial L}{\partial W_{ij}} = \sum_{n=1}^{N}\left(s_i^n d_j^n - \langle s_i d_j\rangle_{p(s,d)}\right)$.
5. **Computational Efficiency:** Infeasible in general case (2^{S+D} sum for partition function Z).

Question 4.2: Empirical KL Minimization & Edge Removal

[Source: 2023–24 Final Exam, Question 2.2.1]
Belief Net $A \to B, A \to C$. Learning parameters by Maximum Likelihood minimizes $KL(Q \parallel P)$ where Q is empirical distribution and P is model distribution.

1. Optimal setting of distribution P and value of $KL(Q \parallel P)$? [5 Marks]
2. Remove edge $A \to B$: impact on optimal KL divergence? [5 Marks]
3. Dataset conditions for equal KL divergence? [5 Marks]

Solution 4.2

Core Concept Summary: Maximum likelihood parameter estimation in Belief Networks is equivalent to minimizing forward KL divergence $KL(Q \parallel P)$ between empirical data Q and model distribution P.

1. **Optimal Setting:** $P^*(A,B,C) = Q(A,B,C) \implies KL(Q \parallel P^*) = \mathbf{0}$.
2. **Edge Removal Impact:** Constrains P to factorizations $P(A)P(B)P(C|A)$. Minimum KL divergence **cannot decrease** ($KL(Q \parallel P^*) \geq 0$).
3. **Identical KL Dataset Condition:** Holds iff A and B are **marginally independent** under empirical data $Q(A,B) = Q(A)Q(B)$.

Question 4.3: Randomized Response Survey Procedure & EM

[Source: 2021–22 Final Exam, Q2.4 & 2023–24 Final Exam, Q2(c)]

1. Graphical model & CPTs. [2 Marks]
2. Truthfulness posterior $P(C_1 = 1 \mid Y = 1)$ for $p_1 = 0.5$. [2 Marks]
3. MLE for $p_1 = 0.5$. [2 Marks]
4. EM updates for unknown (p_1, p_2). [2 Marks]

5. Uniqueness for $D = (\text{yes}, \text{yes}, \text{yes}, \text{no})$? [2 Marks]

Solution 4.3

Core Concept Summary: Randomized response adds hidden coin-flip noise C_1 to protect participant privacy, requiring the Expectation-Maximization (EM) algorithm to estimate unobserved parameters.

1. **Graphical Structure:** $H \to Y \leftarrow C_1$. $P(Y = 1 \mid H, C_1) = C_1 H + (1 - C_1) \cdot 0.5$.
2. **Truthfulness Posterior:** $P(C_1 = 1 \mid Y = 1) = \frac{\mathbf{2p_2}}{\mathbf{2p_2+1}}$.
3. **Closed-Form MLE:** $\hat{p}_2 = \max\left(0, \min\left(1, 2\frac{n_y}{n} - 0.5\right)\right)$.
4. **EM Algorithm Steps:**
   - **E-step:** $\langle C_{1,i} \rangle$ and $\langle H_i \rangle$ posteriors.
   - **M-step:** $p_1^{\text{new}} = \frac{1}{n}\sum_i \langle C_{1,i} \rangle$, $p_2^{\text{new}} = \frac{1}{n}\sum_i \langle H_i \rangle$.
5. **Non-Uniqueness Analysis:** $\hat{\theta} = 0.75 \implies p_1(p_2 - 0.5) = 0.25$. 1D solution manifold. **Not unique**.

Question 4.4: Baum-Welch EM Derivation & Monotonicity Proof

[Source: Classic Sample Exam, Question 2(c)]

1. Derive Baum-Welch E and M steps for transition A and emission B. [6 Marks]
2. Prove log-likelihood is monotonically non-decreasing using free energy bound. [3 Marks]

Solution 4.4

Core Concept Summary: Baum-Welch EM maximizes HMM marginal log-likelihood $\log p(V \mid \theta)$ by iteratively calculating expected transition/emission counts in the E-step and updating A, B matrices in the M-step.

1. **Baum-Welch Updates:** - **E-step:** $\gamma_t^n(i) = \frac{\alpha_t^n(i)\beta_t^n(i)}{p(v_{1:T}^n)}$ and $\xi_t^n(j,i) = \frac{\alpha_{t-1}^n(j)A_{ij}B_{v_t^n,i}\beta_t^n(i)}{p(v_{1:T}^n)}$. - **M-step:** $\hat{A}_{ij} = \frac{\sum_n \sum_t \xi_t^n(j,i)}{\sum_n \sum_t \gamma_{t-1}^n(j)}$ and $\hat{B}_{ki} = \frac{\sum_n \sum_{t:v_t=k} \gamma_t^n(i)}{\sum_n \sum_t \gamma_t^n(i)}$.
2. **Monotonicity Proof:** Log-likelihood decomposition $\log p(V \mid \theta) = \mathcal{B}(q, \theta) + KL(q \parallel p)$. E-step sets $KL = 0$, M-step maximizes $\mathcal{B}(q, \theta)$, ensuring $\log p(V \mid \theta^{\text{new}}) \geq \log p(V \mid \theta^{\text{old}})$.

TOPIC 5: Approximate Inference & Sampling

Question 5.1: Matrix KL Divergence Computation

[Source: 2021–22 Final Exam, Question 3.1]
Given $P(X) = (0.25, 0.25, 0.25, 0.25)^T$, $Q(X) = (0.5, 0.25, 0.125, 0.125)^T$, and matrices P, Q:

$$P = \begin{pmatrix} 0.25 & 0.5 & 0 & 0.2 \\ 0.25 & 0 & 0.5 & 0.3 \\ 0.25 & 0 & 0 & 0.1 \\ 0.25 & 0.5 & 0.5 & 0.4 \end{pmatrix}, \quad Q = \begin{pmatrix} 0.1 & 0 & 0 & 1 \\ 0 & 0 & 0.5 & 0 \\ 0.9 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0.5 & 0 \end{pmatrix}$$

Compute marginals $P(Y), Q(Y)$, and divergences $KL(P(Y) \parallel Q(Y))$ and $KL(Q(Y) \parallel P(Y))$. [5 Marks]

Solution 5.1

Core Concept Summary: KL divergence $KL(P \parallel Q) = \sum P(y) \log \frac{P(y)}{Q(y)}$ is an asymmetric information-theoretic distance measure ($KL(P \parallel Q) \neq KL(Q \parallel P)$).

1. **Marginals Computation:** $P(Y) = [0.2375, 0.2625, 0.0875, 0.4125]^T$, $Q(Y) = [0.1750, 0.0625, 0.5750, 0.1875]^T$.
2. **Forward KL $KL(P \parallel Q)$:**

$$\begin{aligned} KL(P \parallel Q) &= 0.2375 \ln \left(\frac{0.2375}{0.1750}\right) + 0.2625 \ln \left(\frac{0.2625}{0.0625}\right) + 0.0875 \ln \left(\frac{0.0875}{0.5750}\right) + 0.4125 \ln \left(\frac{0.4125}{0.1875}\right) \\ &= \mathbf{0.6097} \text{ nats} \quad (0.8797 \text{ bits}) \end{aligned}$$

3. **Reverse KL $KL(Q \parallel P)$:**

$$\begin{aligned} KL(Q \parallel P) &= 0.1750 \ln \left(\frac{0.1750}{0.2375}\right) + 0.0625 \ln \left(\frac{0.0625}{0.2625}\right) + 0.5750 \ln \left(\frac{0.5750}{0.0875}\right) + 0.1875 \ln \left(\frac{0.1875}{0.4125}\right) \\ &= \mathbf{0.7916} \text{ nats} \quad (1.1420 \text{ bits}) \end{aligned}$$

Question 5.2: Mean-Field Variational Free Energy Derivation

[Source: 2021–22 Final Exam, Q3.2 & 2023–24 Final Exam, Q3(a)]
For 3-node pairwise Markov network $P(X) \propto \prod_{i\sim j} \Phi(X_i, X_j)$ where $\Phi(X_i, X_j) = 2$ if $X_i = X_j$ and 1 if $X_i \neq X_j$, derive explicit free energy $F(q)$ under fully factorized $q(X) = \prod_{i=1}^{3} q_i(X_i)$. [5 Marks]

Solution 5.2

Core Concept Summary: Mean-field variational inference approximates complex joint distributions with fully factorized distributions $q(X) = \prod q_i(X_i)$ by minimizing variational free energy $F(q)$.

$$F(q) = \sum_{i=1}^{3} [\mu_i \log \mu_i + (1-\mu_i)\log(1-\mu_i)] - \log 2 \sum_{(i,j)} [\mu_i \mu_j + (1-\mu_i)(1-\mu_j)]$$

Question 5.3: Mode-Seeking vs Mode-Covering KL Divergence

[Source: Classic Sample Exam, Question 3(a)]
Explain the operational difference between minimizing $KL(P \parallel Q)$ vs $KL(Q \parallel P)$ when approximating bimodal target P with unimodal proposal Q. [5 Marks]

Solution 5.3

Core Concept Summary: Forward KL $KL(P \parallel Q)$ is zero-avoiding (mode-covering), forcing Q to spread across all modes. Reverse KL $KL(Q \parallel P)$ is zero-forcing (mode-seeking), locking Q onto a single mode.

1. **Mode-Seeking ($KL(Q \parallel P)$):** Zero-forcing. Q contracts to cover a single peak of P.
2. **Mode-Covering ($KL(P \parallel Q)$):** Zero-avoiding. Q spreads wide to cover all modes of P.

Question 5.4: Rejection Sampling on Lattices

[Source: 2021–22 Final Exam, Q3(b) & 2023–24 Final Exam, Q3(b)]
Compute exact average acceptance rates $P(\text{accept}) = Z/2^{MN}$ for:

1. $M = 2, N = 1$: $Z = 2 + 2e^{-1} \approx 2.7358 \implies P(\text{accept}) = \mathbf{0.6839}$ (68.39%).
2. $M = 2, N = 2$: $Z = 2 + 12e^{-2} + 2e^{-4} \approx 3.6606 \implies P(\text{accept}) = \mathbf{0.2288}$ (22.88%).

Question 5.5: MCMC, Gibbs & Metropolis-Hastings

[Source: 2021–22 Final Exam, Q3(c), 2023–24 Final Exam, Q3(c) & Classic Sample Exam, Q3(b)]
Detailed balance ($P(x)T(x'|x) = P(x')T(x|x')$) is sufficient for stationary distribution P, but not necessary. Counterexample: Lazy cyclic 3-state chain $T = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \end{pmatrix}$ with $P = [1/3, 1/3, 1/3]^T$. $PT = P$ holds, but $P(1)T(2|1) = 1/6 \neq 0 = P(2)T(1|2)$. Detailed balance is violated.

APPENDIX: Standard Formulas & Foundational Identities

1. Chain Rule of Probability: Step-by-Step Derivation

The Chain Rule permits factorizing any joint probability distribution over N random variables into a product of conditional probabilities.

Fundamental Definition (2 Variables, $N = 2$): For any two random variables x_1 and x_2, the definition of conditional probability is:

$$p(x_2 \mid x_1) = \frac{p(x_1, x_2)}{p(x_1)} \implies p(x_1, x_2) = p(x_1)\, p(x_2 \mid x_1)$$

Complete Step-by-Step Derivation for 3 Variables ($N = 3$): We wish to derive the joint probability $p(x_1, x_2, x_3)$.

1. **Step 1 (First Factorization)**: Treat the pair (x_1, x_2) as a single composite variable $Y = (x_1, x_2)$. Apply the 2-variable definition between Y and x_3:

$$p(x_1, x_2, x_3) = p(Y, x_3) = p(Y)\, p(x_3 \mid Y) = p(x_1, x_2)\, p(x_3 \mid x_1, x_2)$$

2. **Step 2 (Second Factorization)**: Apply the 2-variable definition to expand the marginal term $p(x_1, x_2)$:

$$p(x_1, x_2) = p(x_1)\, p(x_2 \mid x_1)$$

3. **Step 3 (Substitution & Final Equation)**: Substitute $p(x_1, x_2) = p(x_1)\, p(x_2 \mid x_1)$ from Step 2 into the Step 1 equation:

$$\begin{aligned} p(x_1, x_2, x_3) &= [p(x_1)\, p(x_2 \mid x_1)] \cdot p(x_3 \mid x_1, x_2) \\ &= \mathbf{p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_1, x_2)} \end{aligned}$$

General N Variables ($N \geq 2$): By repeating this substitution recursively for N variables:

$$p(x_1, x_2, \ldots, x_N) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_1, x_2) \cdots p(x_N \mid x_1, \ldots, x_{N-1}) = p(x_1) \prod_{i=2}^{N} p(x_i \mid x_1, \ldots, x_{i-1})$$

2. Conditional Probability Tables (CPTs) & Factor Representation

In a Directed Acyclic Graph (DAG) or Belief Network, every variable node x_i is governed by a **Conditional Probability Table (CPT)**, which specifies the probability distribution of x_i given all possible state configurations of its parent nodes $\mathrm{pa}(x_i)$:

$$p(x_i \mid \mathrm{pa}(x_i))$$

- **Root Nodes** ($\mathrm{pa}(x_i) = \emptyset$): The CPT reduces to the prior marginal probability distribution $p(x_i)$.
- **Joint Factorization Rule**: The joint probability distribution over all N variables is computed by multiplying all node CPTs together:

$$p(x_1, x_2, \ldots, x_N) = \prod_{i=1}^{N} p(x_i \mid \mathrm{pa}(x_i))$$

- **Junction Tree Assignment**: Every CPT factor $p(x_i \mid \mathrm{pa}(x_i))$ must be assigned to exactly one clique C_k containing $\{x_i\} \cup \mathrm{pa}(x_i)$, forming initial clique potential $\psi(C_k)$.

3. Bayes' Rule & Law of Total Probability

$$p(y \mid x) = \frac{p(x \mid y)p(y)}{p(x)} = \frac{p(x \mid y)p(y)}{\sum_{y'} p(x \mid y')p(y')}$$

3. D-Separation Rules in Belief Networks

Three canonical 3-node graph structures govern conditional independence in DAGs:

1. **Chain** ($A \to B \to C$): Path blocked when conditioning on middle node B ($A \perp\!\!\!\perp C \mid B$). Unblocked when B is unobserved.
2. **Fork** ($A \leftarrow B \to C$): Path blocked when conditioning on common parent B ($A \perp\!\!\!\perp C \mid B$). Unblocked when B is unobserved.
3. **Collider / Immorality** ($A \to B \leftarrow C$): Path blocked when B and its descendants are unobserved ($A \perp\!\!\!\perp C$). Path **unblocked** when conditioning on B or any descendant of B ($A \not\perp\!\!\!\perp C \mid B$).

4. Kullback-Leibler (KL) Divergence

$$KL(P \parallel Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$$

5. Evidence Lower Bound (ELBO) & Jensen's Inequality

$$\log p(x) \geq \mathbb{E}_{q(z|x)}\left[\log \frac{p(x,z)}{q(z \mid x)}\right] = \mathbb{E}_{q(z|x)}[\log p(x \mid z)] - KL(q(z \mid x) \parallel p(z)) \equiv \mathcal{L}_{\text{ELBO}}$$

6. HMM Dynamic Programming Recursions

$$\alpha_t(h_t) = p(v_t \mid h_t) \sum_{h_{t-1}} p(h_t \mid h_{t-1})\alpha_{t-1}(h_{t-1})$$

$$\beta_t(h_t) = \sum_{h_{t+1}} p(v_{t+1} \mid h_{t+1}) p(h_{t+1} \mid h_t)\beta_{t+1}(h_{t+1})$$

7. Junction Tree Representation & Invariant Joint Factorization

A Junction Tree represents a joint probability distribution over maximal cliques C_i and separator sets S_{ij}:

1. **Initialization Rule**: Each original CPT factor $p(x_k \mid \text{pa}(x_k))$ is assigned to **exactly one** clique C_i containing $\{x_k\} \cup \text{pa}(x_k)$, setting separator potentials $\phi(S_{ij}) = 1$. This guarantees that the product of initial clique potentials equals the exact joint distribution:

$$\prod_i \psi(C_i) \equiv p(x_1, x_2, \ldots, x_N)$$

2. **Invariant Factorization Property**: During and after message absorption, the Junction Tree preserves the joint distribution invariant:

$$p(x_1, x_2, \ldots, x_N) = \frac{\prod_i \psi(C_i)^*}{\prod_{ij} \phi(S_{ij})^*}$$

Upon convergence, calibrated clique potentials $\psi(C_i)^*$ equal true clique marginals $p(C_i)$, and separator potentials $\phi(S_{ij})^*$ equal true separator marginals $p(S_{ij})$.